

EpiForecast-MX — Patent Code Bundle / Paquete de Código para Patente

Version: generated by `scripts/build_patent_bundle.sh` · **Repo:** EpiForecast-MX **Scope:** the curated, self-contained subset that constitutes *the invention* — engine code, configuration, orchestration, and tests — with all documentary bloat, clinical data, and model binaries removed.

⚠️ LEGAL NOTICE / AVISO LEGAL (open item — pending counsel)

EN — This bundle currently ships the repository's existing `LICENSE` (MIT), and the source has been **publicly published** (`github.com/.../EpiForecast-MX` , public; `epiforecast.mx` live). An open MIT release is an **enabling public disclosure** and grants an irrevocable use/sell license, which can undermine patent novelty in absolute-novelty jurisdictions (MX, EPO). **This is a governance decision for legal/IP counsel, not a code issue.** Before any patent filing, confirm the license posture (e.g., replace MIT with an "All Rights Reserved / Confidential" notice aligned across `LICENSE` , `pyproject.toml` , and `README.md`) and audit the disclosure timeline against the 12-month grace period (US 35 U.S.C. §102(b)(1); MX). See the audit report `reports/AUDITORIA_PATENTE_CODIGO.md` , §4.

ES — Este paquete incluye por ahora la `LICENSE` vigente del repositorio (MIT) y el código ya fue **publicado públicamente** (GitHub público; `epiforecast.mx` en vivo). Una publicación MIT abierta es **divulgación habilitante** y concede licencia irrevocable de uso/venta, lo que puede minar la novedad en jurisdicciones de novedad absoluta (MX, EPO). **Es una decisión de gobernanza para el asesor legal/PI, no un defecto de código.** Antes de cualquier trámite, confirmar la postura de licencia (p. ej. sustituir MIT por un aviso "Todos los derechos reservados / Confidencial" alineado en `LICENSE` , `pyproject.toml` y `README.md`) y auditar las fechas de divulgación contra la gracia de 12 meses. Ver el informe de auditoría `reports/AUDITORIA_PATENTE_CODIGO.md` , §4.

1. What is in the bundle / Qué incluye

EN — Only the invention and the evidence it works. **ES** — Solo la invención y la evidencia de que funciona.

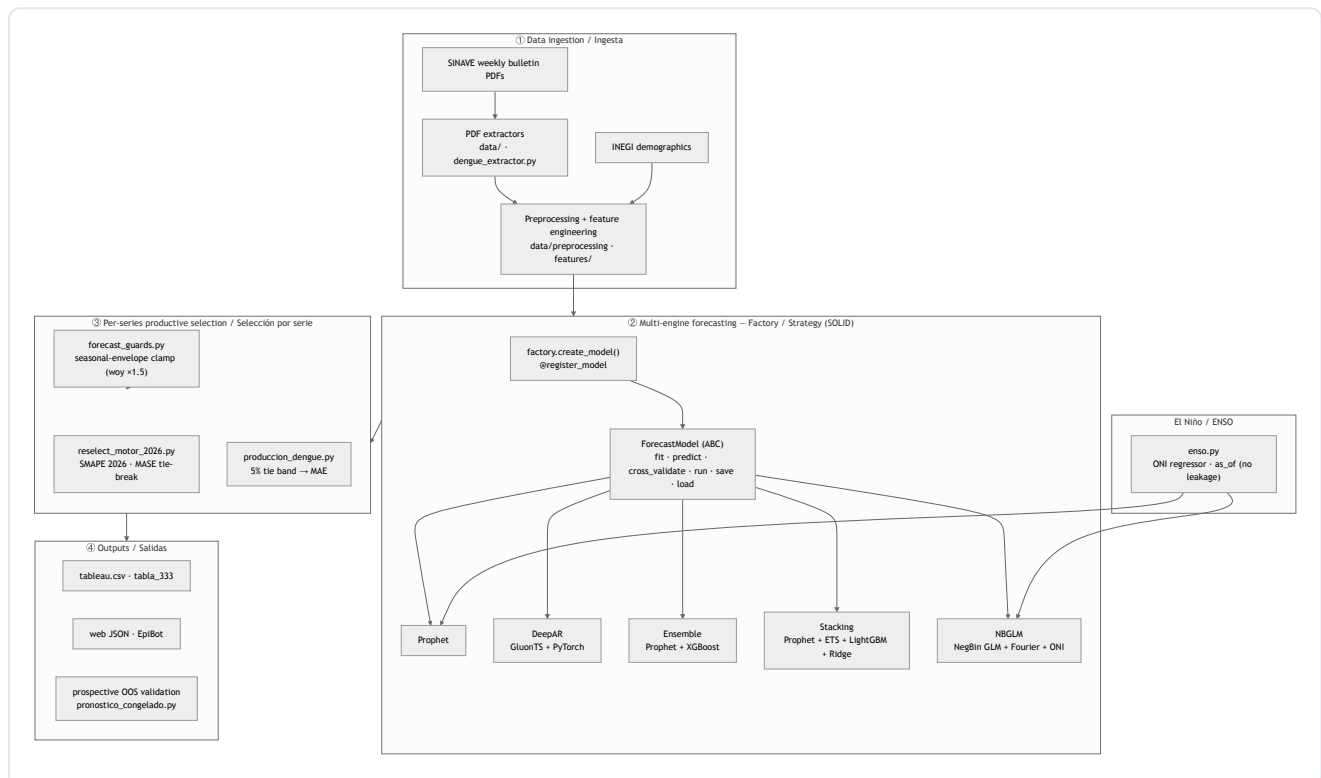
Path	Contents	Why included
<code>src/epiforecast/</code>	Core package (~94 <code>.py</code> , ~15k LOC): <code>models/</code> (Factory + base + 5 engines), <code>data/</code> , <code>features/</code> , <code>evaluation/</code> , <code>visualization/</code> , <code>pipelines/</code> , <code>utils/</code>	The invention itself
<code>scripts/</code>	CLI entry points / orchestration (one-off recovery scripts excluded)	Reproducible operation
<code>config/</code>	OmegaConf YAML that parameterizes the method	The method is config-driven
<code>epi_modules/</code>	Interactive EPI console (peripheral)	Operator tooling
<code>tests/</code>	~915 tests in 56 files	Evidence of correctness
<code>docs/model_cards/</code> , <code>docs/research/INFORME_ARQUITECTURA_MULTIMODELO.md</code>	Engine model cards + architecture report	Technical documentation
<code>pyproject.toml</code> , <code>requirements.txt</code> , <code>Makefile</code> , <code>README.md</code> , <code>.pre-commit-config.yaml</code> , <code>.python-version</code> , <code>epi.py</code> , <code>LICENSE</code>	Build, deps, quality contract, license	Buildable from clean clone

2. What is deliberately excluded / Qué se excluye a propósito

notebooks/ (~~118 MB, base64 embedded outputs~~) — references/ (~213 MB academic PDFs/photos) · reports/ (~134 MB, except the audit report) · data/ and models/ (clinical IMSS data + .pkl binaries, private S3/DVC — **confidential, not claimable**) · aws/ SageMaker infra (optional; contains a hardcoded AWS account id) · CLAUDE.md , GEMINI.md , .epi_history.json (internal AI config / personal history) · Congresos/ , Stories/ (papers under double-blind review) · checkpoints/ , mlruns/ , lightning_logs/ (regenerable artifacts).

The repository is ~5.5 GB on disk; **the invention is ~2.5 MB**. Excluding the documentary bloat is what makes the deliverable reviewable.

3. Architecture / Arquitectura



► ASCII fallback (no Mermaid renderer)

4. Patentable IP map / Mapa de IP patentable

Mapped to `file:function`, ordered by remaining defensibility (some strong candidates are **already partially disclosed** via the MICAI paper and the live site — see the audit report §5). / Mapeado a `archivo:función`, ordenado por defensibilidad restante.

#	Contribution / Contribución	Location
1	Seasonal-envelope clamp by epidemiological week (caps each forecast week at that ISO-week historical max $\times 1.5$, global-max fallback) — engine-agnostic plausibility guard; the <i>algorithm</i> is undisclosed	models/forecast_guards.py (clamp_seasonal_envelope)
2	Deployable ENSO/ONI regressor with damped-persistence future ONI and an as_of cutoff that prevents climate leakage in backtests	data/enso.py
3	Multi-year NBGLM projection (freeze_trend , trend_anchor_weeks , degenerate-series constant fallback)	models/nbglm/model.py
4	Per-series engine selection / tie-break rules (5% SMAPE tie band broken by MAE; "if it's zero, it's zero"; noisy→Ensemble fallback) — the <i>general idea</i> is disclosed; claim the concrete thresholds	scripts/produccion_dengue.py , scripts/reselect_motor_2026.py
5	Frozen-forecast prospective OOS validation protocol	scripts/pronostico_congelado.py

Not patentable (context only): the Factory/registry pattern (factory.py) and ForecastModel interface (base.py) are GoF prior art; the SINAVE PDF extractor is novel in specifics but likely obvious.

5. Build & reproduce / Construir y reproducir

```
# Rebuild this curated bundle from the repository root:
bash scripts/build_patent_bundle.sh
# → dist/patent_bundle/ + dist/epiforecast-patent-bundle.tar.gz + .sha256

# From inside the bundle (clean clone):
python -m venv .venv && . .venv/bin/activate
pip install -e ".[dev]"          # flit-built package `epiforecast`
make quality                    # ruff + mypy (strict) + pytest
python -m scripts.entrena       # train per config/base.yaml
python -m scripts.predice       # forecast
```

Reproducibility caveat / Salvedad: the clinical IMSS data and .pkl model artifacts live in a **private** S3/DVC remote and are **not** in this bundle. To make the deliverable fully self-contained, attach a frozen data + model subset (or the source SINAVE PDFs + an end-to-end regeneration script) with documented checksums. See audit report §1 and §3.

6. Quality gate (measured at audit time) / Calidad medida

- **Ruff** check . : **0 errors**.
- **Mypy** (strict = true): **0 errors** across 94 source files (full return typing).
- **Pytest**: **~915 tests pass**, 0 failures.
- **Coverage**: **70%** (at the gate; core prophet/model.py at 55% — raise before freezing the snapshot).
- **SRP (≤ 300 lines/module)**: 15 modules exceed it, concentrated in visualization/reporting (peripheral), not in the claimable logic; only deepar/model.py is the documented exception.

Full detail, severities, and the remediation checklist: reports/AUDITORIA_PATENTE_CODIGO.md .